Software Requirements Specification

for

Clinic Management System

Version 1.0 (Draft for review)

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Status: pending client confirmation of the TBD items in Appendix C

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Revision History

Name	Date	Reason For Changes	Version
Team B	11/10/2026	Initial draft from client meeting notes of 05/10/2026	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Clinic Management System (CMS), Version 1.0. It is a reference for the client, developers, testers and project guides. It describes the complete system covering the Patient, Receptionist, Doctor and Management/Administrator roles, and follows IEEE Std 830-1998 as in the supplied template.

1.2 Document Conventions

Functional requirements are identified by the prefix "FR-" and a two-digit number (e.g., FR-01) and are written as "The system shall ...". Each feature has a priority of High, Medium or Low. Higher-level features inherit the priority of their most critical requirement. Unresolved decisions are marked TBD-n and listed in Appendix C; no unresolved value has been guessed.

1.3 Intended Audience and Reading Suggestions

This document is intended for the client (clinic management and reception), developers, testers, database administrators and evaluators. Readers should begin with Section 2 for context, then read Section 3 (interfaces) and Section 4 (features and functional requirements). Sections 5 and 6 give non-functional and other requirements; the appendices contain the glossary, analysis models and the TBD list.

1.4 Product Scope

The CMS is a web-based application that automates appointment booking and clinic operations for a clinic with six departments. Patients raise appointment requests with a booking charge; the receptionist approves them according to doctor availability; doctors' shifts and holidays are managed; patient history, symptoms and digital reports are kept; encrypted medical files can be uploaded by the receptionist without being read; and notifications and real-time cancellation updates are provided. The system aims to reduce no-shows and manual effort. No payment gateway is integrated, and online payment, billing and telephony are out of scope. A mobile app was mentioned in early notes and is unconfirmed (TBD-5).

1.5 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Client meeting notes (Minutes of Meeting) on the Clinic Management System, 05/10/2026.
- Team B requirement-gathering notes, October 2026.
- Supplied SRS templates and the Digital Bank Management System SRS (format reference only).

2. Overall Description

2.1 Product Perspective

The CMS is a new, self-contained web-based system that replaces manual, phone-based booking. It follows a client-server architecture: users work in a web browser, and a central server holds the database and encrypted file storage. The clinic has no server room, so hosting is external (cloud preferred; local server alternative: TBD-12). Different interfaces are provided for each role based on permissions.

2.2 Product Functions

Patient Functions

- Register, log in and log out
- Fill medical history; request an appointment and see the booking charge
- Cancel an appointment before the cut-off; view appointments and notifications

Receptionist Functions

- Approve or reject requests; book on behalf of patients; maintain patient records
- Edit doctor schedules and holidays; record symptoms (also offline)
- Upload encrypted medical files (metadata only visible); handle cancellations and call patients individually

Doctor Functions

- View schedule and mapped patients; view history, reports and authorised files; add notes

Management/Administrator Functions

- Manage accounts, departments and configuration; edit schedules; generate operational reports; view the audit log

The use case diagram is shown in Figure 1 (Appendix B.1).

2.3 User Classes and Characteristics

User class	Characteristics
Receptionist	Most important user; basic to moderate computer skills; uses a laptop/PC daily; master login (TBD-2).
Patient	Varied computer skills; occasional use.
Doctor	Basic computer skills; intermittent use; sees only mapped patients.
Management/Administrator	Basic skills; authority over accounts, configuration and reports.

2.4 Operating Environment

Component	Specification
Operating System	Windows or Linux (staff PCs)
Browser	Current Chrome, Edge or Firefox
Server / hosting	Cloud (AWS discussed) or local server, 50-100 GB - TBD-12
Technology stack	Not confirmed; to be chosen by the team (proposed)

2.5 Design and Implementation Constraints

- No payment gateway shall be used.
- No server room is available; hosting is external.
- Personal and medical data shall be encrypted and protected by role-based access control (TBD-11).
- Delivery timeline is three weeks; delay terms and cost responsibility are contractual matters (TBD-13).
- Real-time updates and prevention of double booking are required.

2.6 User Documentation

An on-screen help section, validation and confirmation messages, an installation guide and a user manual for each role will be delivered. Formats are TBD.

2.7 Assumptions and Dependencies

- Network connectivity in the building is reliable; offline entry covers short outages.
- A requested slot is held while the request is Pending Approval.
- Booking charges are collected outside the system; only their status is recorded.
- The cancellation cut-off of 2 hours before the appointment is the default (TBD-6).
- Doctor and receptionist accounts are created by Management.
- The client supplies departments, doctors, fees and shift timings.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a simple, responsive graphical interface with role-specific dashboards:

- Home, Registration and Login pages
- Patient Dashboard (appointment request, charge display, cancellation, notifications)
- Receptionist Dashboard (pending requests, calendar, availability, patient search, symptom entry, file upload)
- Doctor Dashboard (schedule, mapped patients, history, files, notes)
- Management Dashboard (accounts, departments, configuration, reports, audit log)

The interface shall provide clear navigation, validation messages, confirmation alerts, an offline indicator and a help option.

3.2 Hardware Interfaces

The system shall operate on standard laptops/PCs and patient devices with a keyboard, mouse or touch input, a modern browser and an internet connection. No special hardware is required.

3.3 Software Interfaces

Component	Technology
Browser	Chrome, Edge, Firefox
Database	Relational database (product TBD)
File storage	Encrypted storage for medical files
Notification	In-app; SMS/e-mail only if confirmed (TBD-10)
API format	Not confirmed (proposed: REST, JSON)

3.4 Communications Interfaces

- HTTPS (TLS) for all client-server communication
- Real-time updates to open browser sessions
- Secure transfer of files and notifications

4. System Features

This section organizes the functional requirements by system feature. Each feature lists its description, priority and the associated functional requirements.

4.1 User Registration and Authentication

4.1.1 Description and Priority

Allows patients to register, authenticates all users, enforces role-based access control and supports secure logout. The receptionist uses a general (master) login as requested by the client. Priority: High.

4.1.2 Functional Requirements

FR-01: The system shall allow a patient, or the receptionist on the patient's behalf, to register with full name, date of birth or age, e-mail, primary contact number, password and an optional second contact number, rejecting invalid or duplicate data with clear messages.

FR-02: The system shall authenticate users with username/e-mail and password before granting access and shall show a generic error for invalid credentials.

FR-03: The system shall provide the roles Patient, Receptionist, Doctor and Management/Administrator and shall enforce their permissions on the server for every request, denying and logging unauthorised access.

FR-04: The system shall end a session after the configured inactivity period (TBD-3) and on logout.

FR-05: The system shall provide the receptionist login required by the client (shared master or individual accounts: TBD-2) and shall record the account used for every action.

4.2 Appointment Request and Booking

4.2.1 Description and Priority

A patient raises an appointment request; the receptionist approves or rejects it according to doctor availability. A booking charge applies; no payment gateway is used. Priority: High.

4.2.2 Functional Requirements

FR-06: The system shall allow a patient to submit an appointment request with department, doctor, date, slot and reason.

FR-07: The system shall offer only slots within a doctor's shift that are not holidays and not already held.

FR-08: The system shall create each valid request as Pending Approval and hold the slot.

FR-09: The system shall let the receptionist approve a request (Confirmed) or reject it with a reason (Rejected, slot released), and shall notify the patient.

FR-10: The system shall prevent double booking of a doctor and slot, including simultaneous requests.

FR-11: The system shall let the receptionist book on behalf of a patient who enquires by phone or in person.

FR-12: The system shall show the booking charge (a configurable percentage, 10% or 15% discussed: TBD-4) before submission and record its status without any payment gateway.

FR-13: The system shall require a new patient's diagnosis history before the first appointment is confirmed.

4.3 Appointment Cancellation

4.3.1 Description and Priority

Supports cancellation before two hours of the appointment, emergency cancellation with minimal refundable charges, and real-time updates. Priority: High.

4.3.2 Functional Requirements

FR-14: The system shall allow a patient to cancel a Pending or Confirmed appointment until 2 hours before its start time (interpretation: TBD-6).

FR-15: The system shall disable cancellation after the cut-off and show a message.

FR-16: The system shall let the receptionist mark a cancellation as an emergency and record the reason; the charge and refund follow the rule in TBD-6 (no refund is executed by the system).

FR-17: The system shall release the slot and update calendars and dashboards in real time on cancellation.

FR-18: The system shall allow marking a confirmed appointment as No-show; any charge is TBD-7.

4.4 Doctor Scheduling and Availability

4.4.1 Description and Priority

Maintains doctor profiles, monthly shifts (Shift 1 and 2), pre-planned holidays and the availability dashboard, editable by the receptionist and authorised management. Priority: High.

4.4.2 Functional Requirements

FR-19: The system shall store each doctor's name, specialty and department.

FR-20: The system shall let authorised users define monthly shifts and planned holidays, and shall treat those periods as bookable or not accordingly.

FR-21: The system shall allow only the receptionist and authorised Management users to edit schedules and shall warn when a change affects confirmed appointments.

FR-22: The system shall provide an availability dashboard and appointment calendar.

FR-23: The system shall record the room assigned to each shift (number of rooms and minimum doctor rule: TBD-8).

4.5 Patient Records, Symptoms and Offline Updates

4.5.1 Description and Priority

Maintains patient profiles, medical history, symptoms and doctor-patient mapping. The receptionist can update symptoms offline. Priority: Medium.

4.5.2 Functional Requirements

FR-24: The system shall store patient profile, medical history and digital reports in the patient record history with the user and time of each change.

FR-25: The system shall let the receptionist record symptoms for a patient.

FR-26: The system shall let the receptionist continue symptom entry offline, keeping entries locally until the server acknowledges them.

FR-27: The system shall synchronise offline entries automatically without duplicates, flag conflicts for review, and show unsent entries after a failed synchronisation (policy: TBD-9).

FR-28: The system shall map patients to doctors and show a doctor only the patients mapped to that doctor.

4.6 Encrypted Medical File Handling

4.6.1 Description and Priority

The receptionist may upload and acknowledge an encrypted file without reading it; only authorised medical personnel may read the contents. Encryption is combined with role-based access and auditing. Priority: High.

4.6.2 Functional Requirements

FR-29: The system shall store medical files encrypted and transmit them only over encrypted connections (algorithm and keys: TBD-11).

FR-30: The system shall let the receptionist upload a file and see only its metadata (patient, name, type, size, time, status); the receptionist shall not open, preview or download its contents.

FR-31: The system shall let only a doctor mapped to the patient view the decrypted contents.

FR-32: The system shall deny and log every other attempt to read file contents, and log every upload and access.

4.7 Notifications, Reports and Administration

4.7.1 Description and Priority

Notifies users of events, keeps digital reports, supports management reports, audit logs and configuration. Priority: Medium.

4.7.2 Functional Requirements

FR-33: The system shall create in-app notifications when a request is received, approved, rejected or cancelled and when a schedule change affects an appointment (other channels: TBD-10).

FR-34: The system shall reflect status and availability changes on open screens in real time (delay target: TBD-13).

FR-35: The system shall provide digital reports in patient history, filterable by date and doctor, according to role permissions.

FR-36: The system shall let Management generate reports of appointments, cancellations and no-shows.

FR-37: The system shall log logins, record access, denied attempts, appointment status and configuration changes, readable only by Management and not editable.

FR-38: The system shall let Management maintain departments (six at launch), rooms, booking-charge percentage and cancellation cut-off.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support at least 100 concurrent users without significant degradation (value from notes; confirmation TBD-13).
- The system shall respond within a limit to be agreed (TBD-13); 3 seconds is proposed.
- The design shall be sized for about 300 patients per day on average (higher on Saturdays and Sundays), about 400 new users per month and six departments; no load is guaranteed until tested.

5.2 Safety Requirements

Booking, cancellation and status changes shall be atomic so that partial or double bookings cannot occur, and data loss shall be minimised through regular backups (frequency: TBD-11).

5.3 Security Requirements

- Passwords shall be stored as salted hashes; personal data and medical files shall be encrypted (TBD-11).
- Role-Based Access Control shall be enforced; receptionists shall not read medical file contents.
- Sessions shall expire after inactivity (TBD-3); security events shall be audited.
- Cloud confidentiality policy (AWS discussed) shall be accepted by the client; no regulatory compliance is claimed.

5.4 Software Quality Attributes

Reliability

- Transactions shall be consistent; offline entries shall not be lost.

Availability

- The system shall be available 24x7, except for scheduled maintenance (uptime target: TBD-11).

Usability

- The interface shall be simple for staff with basic skills; error messages shall be clear.

Scalability

- The system shall support more departments, doctors and users without major redesign.

Maintainability

- Modular, documented code; charges and cut-offs configurable without code changes.

Portability

- The web application shall run on Windows and Linux in modern browsers.

5.5 Business Rules

- A booking is confirmed only after receptionist approval.
- The booking charge is a configurable percentage; no payment gateway is used.
- Patients may cancel only before the cut-off; emergency cancellations carry a minimal refundable charge.
- No doctor, room or slot may be booked twice; no booking on holidays.
- A new patient must provide diagnosis history before the first confirmed appointment.
- Receptionists can see that a file exists but cannot read it; doctors see only mapped patients.
- Only the receptionist and authorised Management edit schedules; only Management changes configuration and views the audit log.

6. Other Requirements

6.1 Database Requirements

The CMS shall maintain the following tables (proposed logical design). The database shall enforce primary keys, foreign keys and referential integrity.

Table Name	Description
User_Account	Login credentials and roles
Patient	Patient personal information (two contact numbers)
Doctor	Doctor profile and department
Department	Clinic departments (six)
Room	Consultation rooms
Doctor_Schedule	Monthly shifts per doctor and room
Doctor_Leave	Planned holidays and unavailable periods
Appointment	Appointments, status, booking charge, cancellation data
Medical_Record	Diagnosis history
Symptom_Entry	Symptoms including offline entries
Medical_File	Encrypted file metadata and references (contents in encrypted storage)
Report	Digital reports
Doctor_Patient_Map	Doctor-patient mapping
Notification	System notifications
Audit_Log	User activity log
System_Config	Configurable parameters

6.2 Database Constraints

- Each table shall have a primary key and foreign keys shall maintain referential integrity.
- At most one active (Pending or Confirmed) appointment per doctor and slot.
- Patient e-mail shall be unique; offline symptom entries carry a unique client entry identifier.
- Status changes shall be atomic and follow the valid appointment states.
- Medical file contents shall never be stored in unrestricted form.

6.3 Testing Strategy

The system shall undergo unit, integration, system and user acceptance testing. Cases that depend on an undecided rule depend on the stated TBD item.

Test Case ID	Test Scenario	Expected Result
TC-01	Login with valid credentials	User logs in successfully
TC-02	Login with invalid password	Generic error displayed; attempt logged
TC-03	Request appointment with available doctor	Appointment Pending Approval
TC-04	Request when no slot available	Request rejected; alternatives shown
TC-05	Two requests for the same slot	Only one active appointment
TC-06	Receptionist approves / rejects	Status updated; patient notified
TC-07	Cancel before / after cut-off	Cancelled and slot released / refused
TC-08	Receptionist opens an uploaded file	Access denied; metadata only; logged
TC-09	Mapped doctor opens file	Contents shown; access logged
TC-10	Offline symptom entry then reconnect	Entry stored once after sync

6.4 Future Scope

- Mobile application for patients (TBD-5)
- SMS and e-mail notifications (TBD-10)
- Online payment integration (outside current scope)
- Multi-factor authentication
- Analytics dashboard

Appendix A: Glossary

Term	Description
SRS	Software Requirements Specification
CMS	Clinic Management System
RBAC	Role-Based Access Control
TBD	To Be Determined
Patient	Registered user who requests appointments
Receptionist	Staff member who approves bookings and maintains records
Appointment	Booking of a doctor slot
Booking charge	Percentage of the fee charged at booking
No-show	Patient who does not attend a confirmed appointment
Encrypted file	Medical file whose contents are unreadable without authorised access
Audit Log	Record of system activities performed by users

Appendix B: Analysis Models

B.1 Use Case Diagram

Figure 1 illustrates the interaction between Patient, Receptionist, Doctor and Management/Administrator and the system.

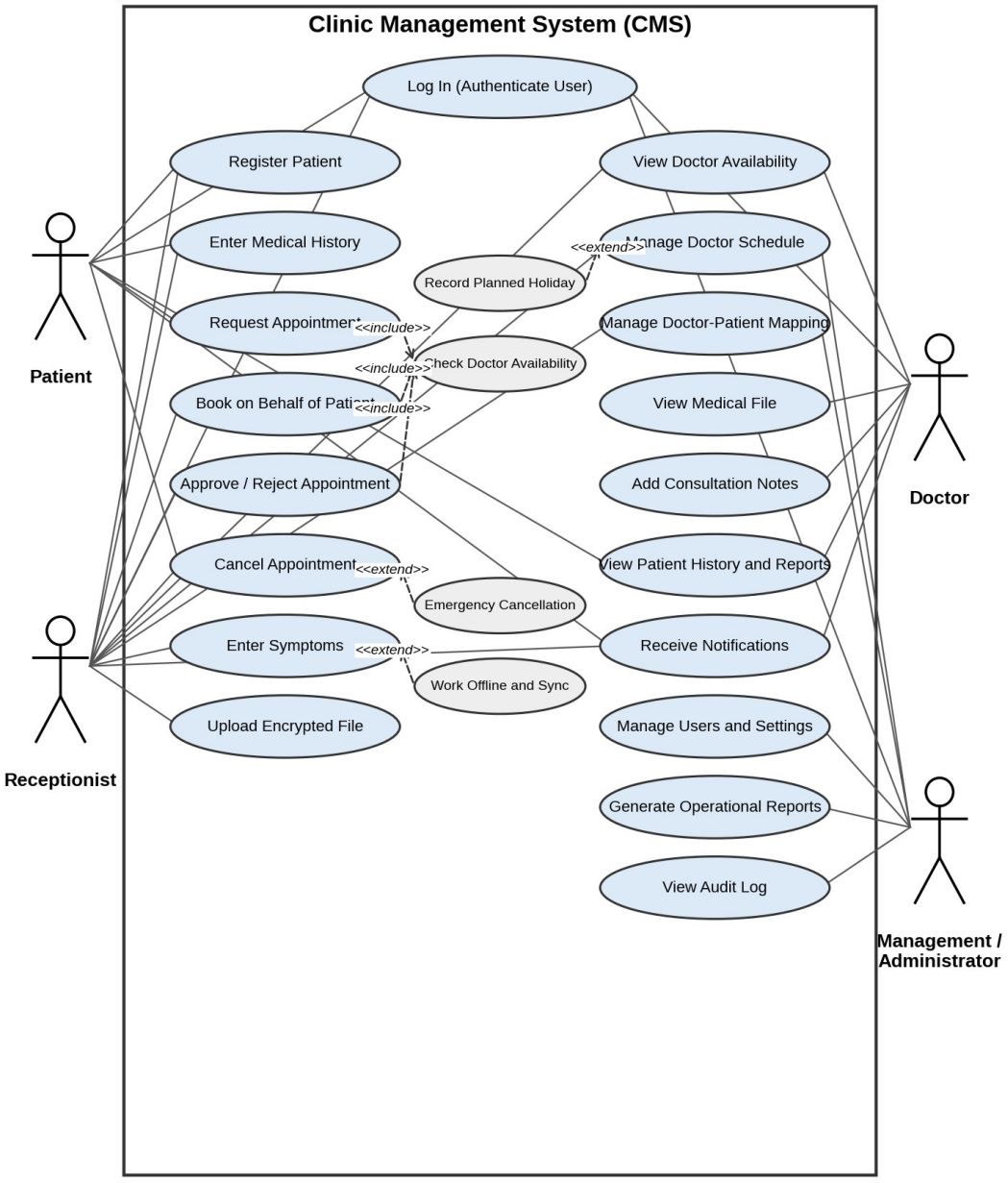


Figure 1: Use Case Diagram

B.2 Class Diagram

Figure 2 represents the static structure: User and its subclasses, Appointment, DoctorSchedule, MedicalRecord, MedicalFile and related classes.

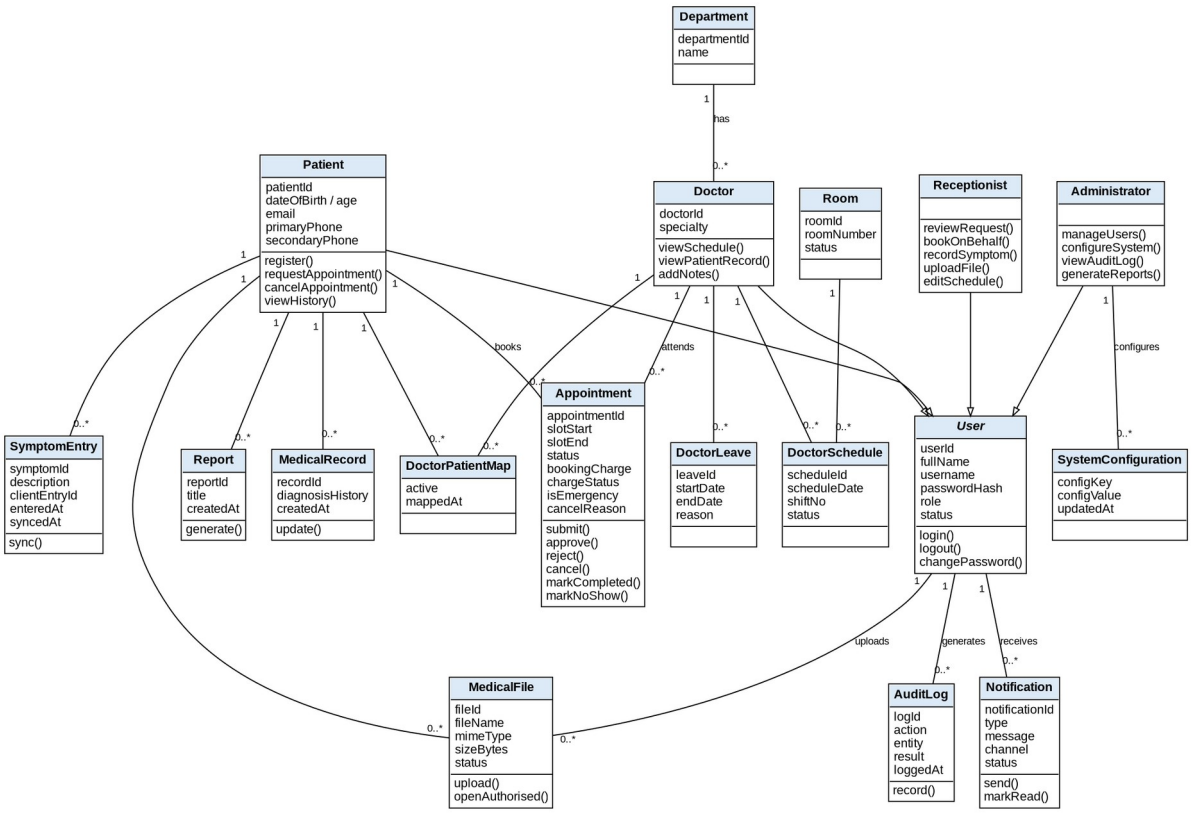


Figure 2: Class Diagram

B.3 Activity Diagram

Figure 3 illustrates the booking workflow from request to notification, with branches for invalid data, unavailable slot, conflict and rejection.

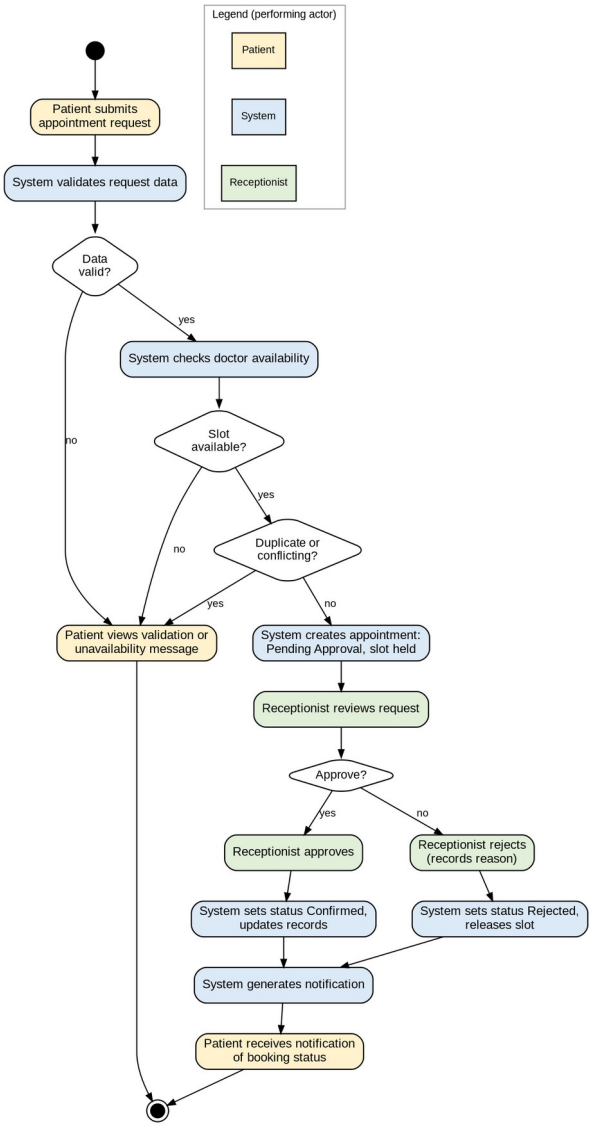


Figure 3: Activity Diagram - Appointment Booking

B.4 Sequence Diagram

Figure 4 shows the interaction between Patient, Web Interface, Application Service, Booking Service, Database, Receptionist and Notification Service during request and approval.

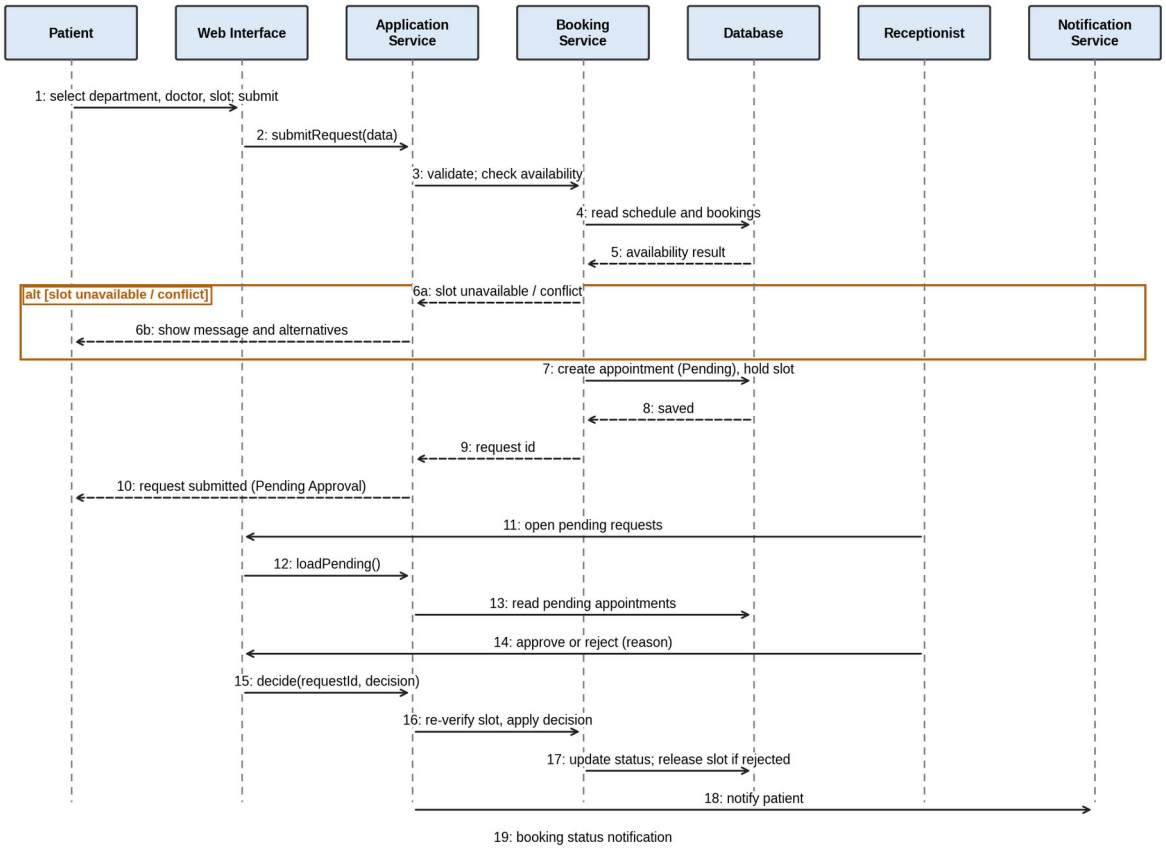


Figure 4: Sequence Diagram - Appointment Request and Approval

B.5 Entity Relationship (ER) Diagram

Figure 5 illustrates the relationships among the proposed database entities (requires client confirmation).

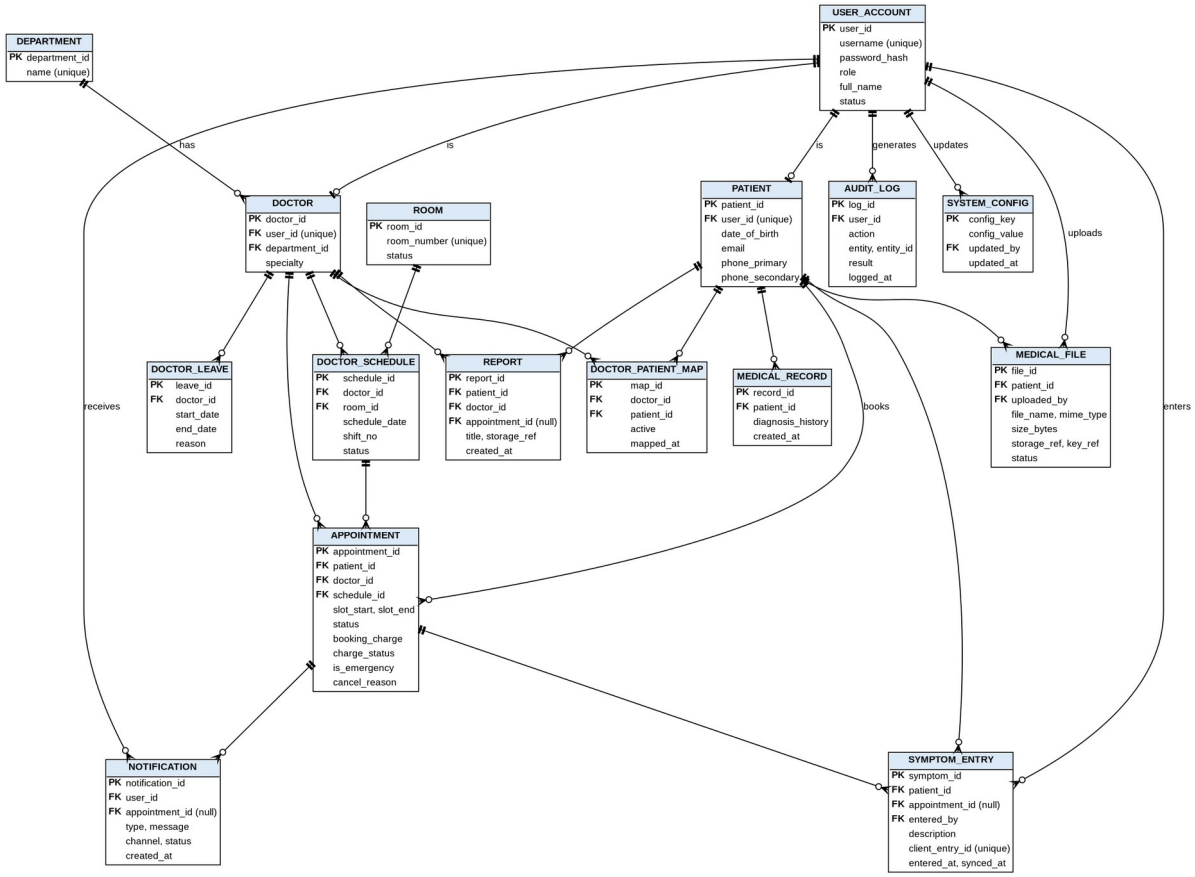


Figure 5: Entity-Relationship Diagram

B.6 Data Flow Diagrams (DFD)

Level 0 shows the complete system as a single process with its external entities. Level 1 divides it into User and Access Management, Patient Management, Doctor Scheduling, Appointment Processing, Medical Record Handling, Notifications and Reports, and Administration and Audit. Level 2 decomposes Appointment Processing.

Level 0

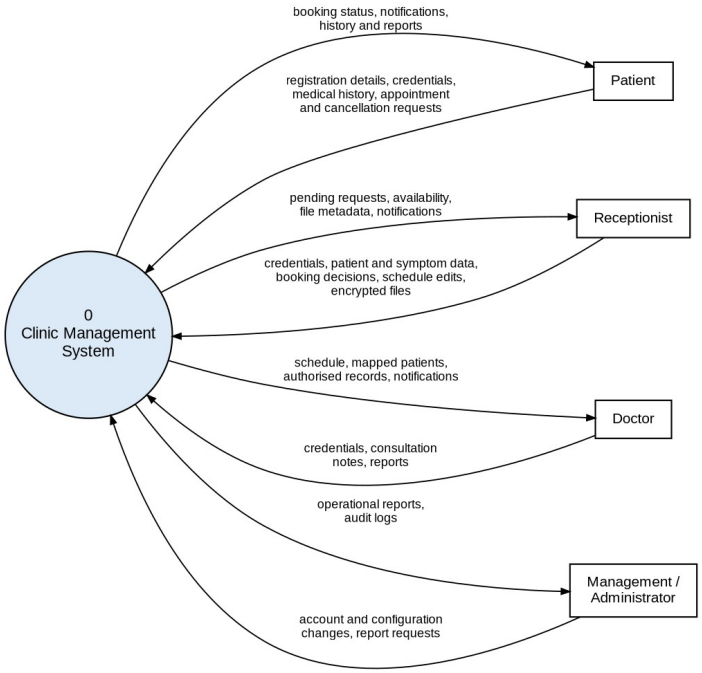


Figure 6: DFD Level 0

Level 1

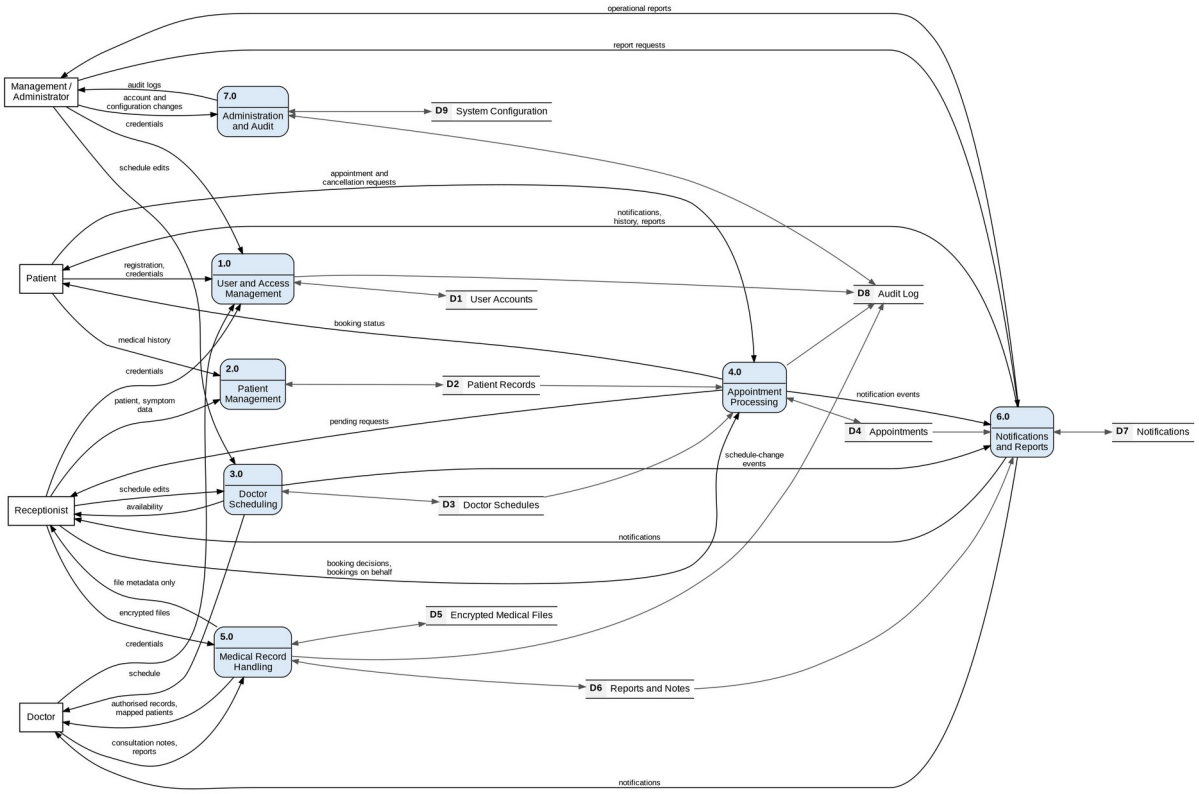


Figure 7: DFD Level 1

Level 2

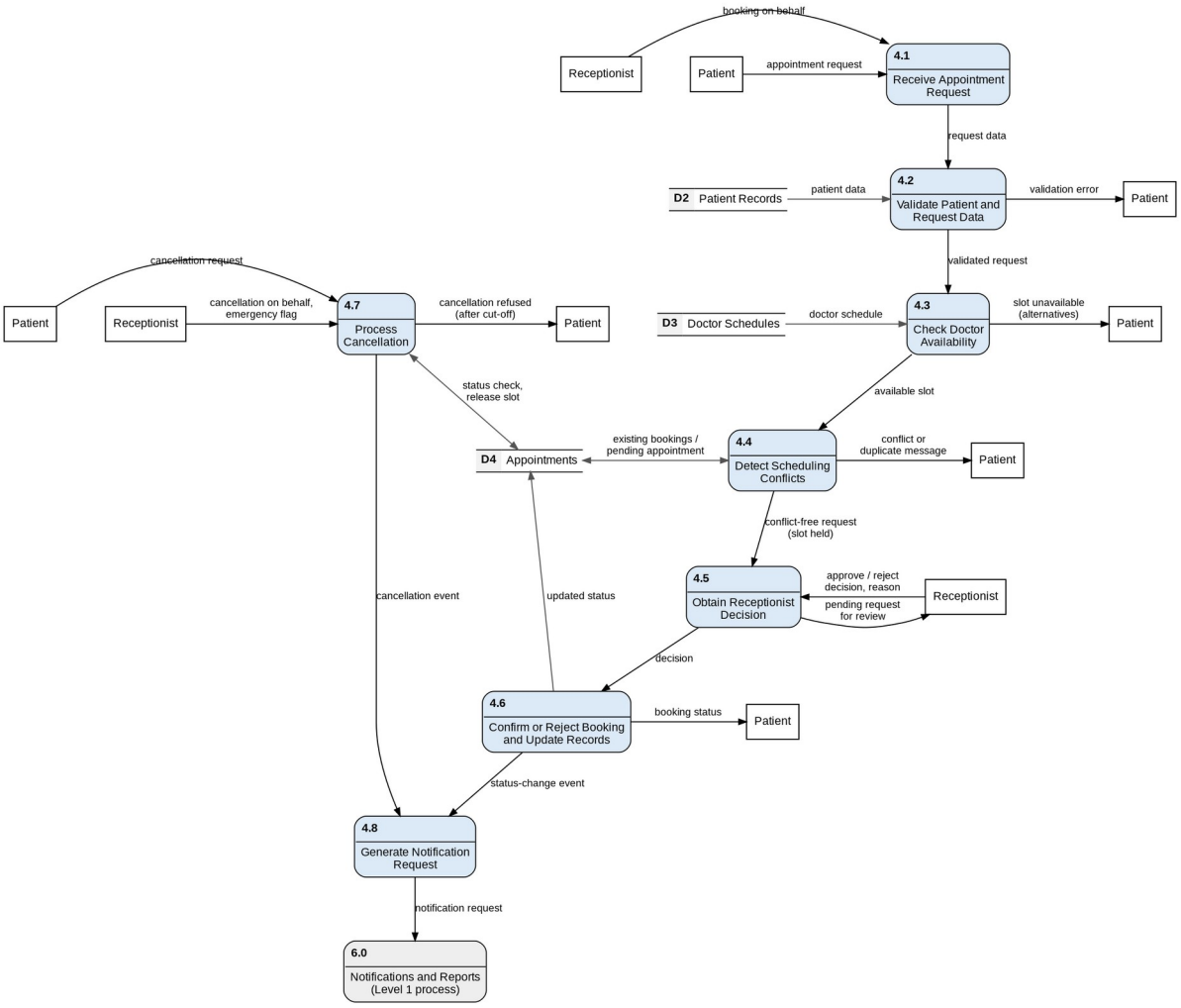


Figure 8: DFD Level 2 - Appointment Processing

B.7 Use Case Specifications

ID	Use Case	Primary Actor	Description
UC-01	Patient Registration	Patient	Allows a new patient to register in the system.
UC-02	User Login	All roles	Authenticates users and grants access according to role.
UC-03	Request Appointment	Patient	Patient requests a slot and sees the booking charge; request becomes Pending Approval.
UC-04	Approve / Reject Appointment	Receptionist	Reviews pending requests and confirms or rejects them based on availability.
UC-05	Book on Behalf of Patient	Receptionist	Creates a booking for a patient who enquires by phone or in person.
UC-06	Cancel Appointment	Patient, Receptionist	Cancels before the cut-off; supports emergency cancellation; releases the slot.
UC-07	Manage Doctor Schedule	Receptionist, Management	Defines monthly shifts and holidays and updates the availability dashboard.
UC-08	Enter Symptoms (offline)	Receptionist	Records symptoms, also offline, and synchronises without duplicates.
UC-09	Upload Encrypted File	Receptionist	Uploads a file; only metadata is visible to the receptionist.
UC-10	View Medical File	Doctor	Mapped doctor views decrypted contents; access is logged.

ID	Use Case	Primary Actor	Description
UC-11	View History and Reports	Doctor, Patient	Views appointments, notes and reports according to role.
UC-12	Manage Users and Settings	Management	Manages accounts, departments, configuration and views the audit log.

Appendix C: To Be Determined List

The following items remain outstanding and require client confirmation before this SRS can be considered complete:

ID	Open item
TBD-1	Budget and cost figures conflict: about Rs 70,000 (budget), Rs 7 lakh (total cost), Rs 70,000-80,000 (linked to recurring charges for 400 new users per month). Meaning and billing period unclear.
TBD-2	Receptionist login: one shared master login or individual accounts (and number of accounts).
TBD-3	Session timeout, password rules and lockout threshold.
TBD-4	Booking charge percentage (10% or 15% discussed), basis and how it is collected and recorded.
TBD-5	Mobile application scope (early notes) versus web baseline; whether the second contact number is mandatory.
TBD-6	Cancellation rules: meaning of the 2-hour cut-off, emergency definition, minimal refundable charge and refund handling.
TBD-7	No-show policy and any charge.
TBD-8	Number of rooms and doctors; meaning of "minimum 2 x X doctors".
TBD-9	Offline symptom synchronisation, conflict policy and local data protection.
TBD-10	Notification channels (in-app, SMS, e-mail) and reminder timing.
TBD-11	Encryption algorithm, key management, retention period, backup frequency and availability target.
TBD-12	Hosting: cloud (AWS Rs 5,000-6,000 per month) versus recurring estimate Rs 2,000-5,000 per month versus local server 50-100 GB with unknown one-time cost.
TBD-13	Three-week delivery terms and delay cost responsibility; meaning of 300 daily patients versus 400 new users per month; maximum concurrent users and response time.